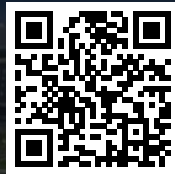


JUMPSTART 2026 • DAY 1



How systems behave

I will post today's slides
and other material here.

gsathish.github.io/JumpStart/

Choices, models, and consequences.

Take one questionnaire, one 0, and one 1.

Start with questions 1–8.

Write your name on the questionnaire. Oddly specific answers work best.

I will show you the two visual questions next.

VISUAL ROUND A

Who is the person in the portrait?



VISUAL ROUND B

What does each stamp celebrate?



1



2



3

Write your best guesses beside question 10.

FIND YOUR IMPROBABLE TWIN

**Walk. Compare answers.
One point for every match.**

Meet at least four people. A person can match more than once.

| When time is up, who found the most matches?

VISUAL ROUND A • REVEAL



scene capture via The Anne Boleyn Files

Anne Boleyn.

Her portrait appears on
Hogwarts' Grand Staircase
in *Harry Potter and the
Philosopher's Stone*.

VISUAL ROUND B • REVEAL

Maxwell. Newton. Einstein.



1 • Maxwell's law



2 • Newton's law

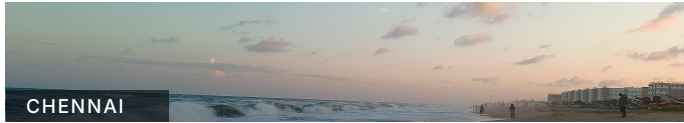


3 • Mass–energy equivalence

Nicaragua, 1971 • "Mathematical Equations Which Changed the World"

Source: MacTutor History of Mathematics, University of St Andrews

WHO I AM



SATHISH GOPALAKRISHNAN
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MY RESEARCH • 2021–2026 • SOURCE: DBLP

scheduling **real-time** **edge-cloud**

ML ensembles **resilience** **serverless**

fault tolerance **distributed systems** **Internet of Things**

resource management explainable AI **sustainable computing**

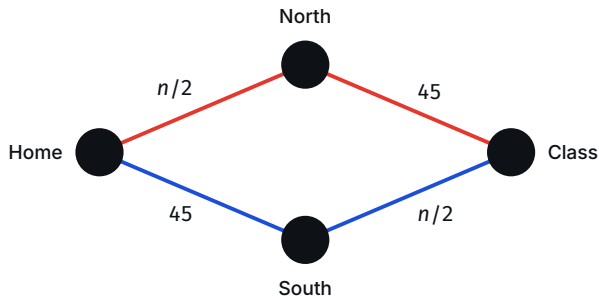
training data performance prediction adaptivity repair education

01

Choose a route

You want to get to class. So does everyone else.

THE MAP



Every number is minutes.

On a road marked $n/2$, the time depends on how many people use it.

If 20 people take that road, it costs 10 minutes.

YOUR FIRST COMMUTE

- 1 Choose privately. Do not follow your neighbour.
- 2 1 is the upper route. 0 is the lower route.
- 3 When I count down, hold up one numeral.

| You want the shortest trip for yourself.

COUNT THE ROOM

Keep your numeral up.

Each section counts 1s and 0s. The four section counters bring me their totals.

1

0

WHAT DID OUR CHOICES COST?

upper route

$$45 + U/2$$

lower route

$$45 + L/2$$

| If one route is slower, would anyone on it switch?

WHERE THE SWITCHING STOPS

34_{upper}

34_{lower}

62 minutes

Either route costs the same. One person switching would make the new route slower.

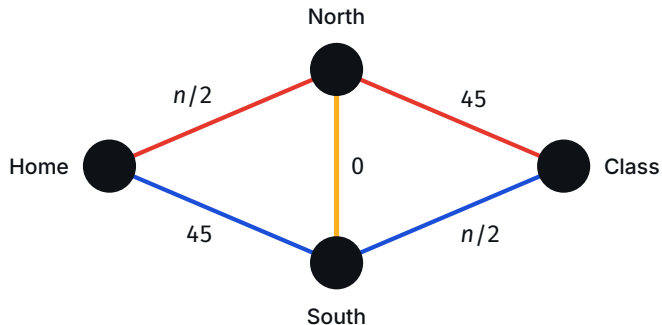
| We have reached an equilibrium.

02

A new road

It costs no time at all.

THE CITY ADDS A SHORTCUT



A zero-minute road.

It runs from North to South. Nothing else on the map changes.

Would you use the new road?

**Do the calculation before you
vote.**

THREE POSSIBLE ROUTES

- 1 sheet: upper route.
- 0 sheet: lower route.
- Both sheets: upper road, shortcut, lower road.

Which route gives you the shortest trip, whatever everyone else does?

THE PRIVATE CALCULATION

On the shortcut route, both changing roads cost at most 34 minutes.

Each ordinary route contains a fixed 45-minute road. The shortcut replaces that 45 with something smaller.

| For one commuter, the shortcut always looks faster.

VOTE AGAIN

1, 0, or both.

Choose privately. Reveal together. Keep your sheets up while your section counts.

upper (1)

lower (0)

shortcut (both)

USE OUR TALLY

- Upper route: $45 + (U + S)/2$
- Lower route: $45 + (L + S)/2$
- Shortcut route: $34 + S/2$

| Would anyone improve their own trip by switching?

THE NEW EQUILIBRIUM

68 minutes

Everyone takes the shortcut. Both changing roads now cost 34 minutes.

The free road has made every commuter six minutes late.

Nobody chose congestion.

So who caused it?

03

A different question

What is best for all 68 commuters?

USE THE CALCULUS YOU ALREADY KNOW

Let y people use the shortcut.

Split everyone else evenly between the two ordinary routes.

ordinary-route time

$$62 + y/4$$

shortcut time

$$34 + y/2$$

| Write the total travel time for all 68 people. Then minimize it.

TOTAL TRAVEL TIME

$$T(y) = (68 - y)(62 + y/4) + y(34 + y/2)$$

TOTAL TRAVEL TIME

$$T(y) = (68 - y)(62 + y/4) + y(34 + y/2)$$

$$T(y) = 4216 - 11y + y^2/4$$

TOTAL TRAVEL TIME

$$T(y) = (68 - y)(62 + y/4) + y(34 + y/2)$$

$$T(y) = 4216 - 11y + y^2/4$$

$$| \quad T'(y) = -11 + y/2 = 0 \quad \implies \quad y = 22$$

THE CITY CAN DO BETTER

23 upper

22 shortcut

23 lower

average trip

60.2 min

With no shortcut, the average was 62. At the voluntary equilibrium, it is 68.

THERE IS A CATCH

The 46 people on the ordinary routes spend 67.5 minutes travelling.

The 22 people on the shortcut spend 45 minutes. An ordinary-route commuter can switch and save 22 minutes.

The best arrangement for the city will not survive voluntary choice.

**An equilibrium is not
the same as a good
outcome.**

WRITE BEFORE WE TALK

- Is the new road bad, even though it makes a better arrangement possible?
- Is closing a road a loss of freedom if everyone gets home sooner?
- What should an engineer optimize: the average, the worst trip, or fairness?
- When reasonable choices produce a bad system, who is responsible?

**Can a constraint
make everyone freer?**

THE ENGINEER'S PROBLEM

The derivative gave us an answer. It did not choose the question.

- Minimize total time, and 46 people take the slower route.
- Let everyone choose, and all 68 people lose six minutes.
- Close the shortcut, and the word freedom becomes awkward.

WHAT MATHEMATICS CAN DO

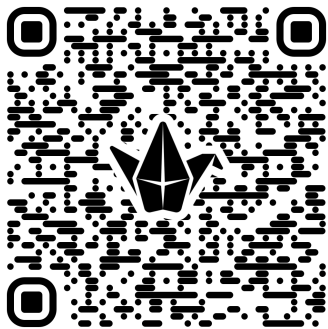
It can tell us what follows from our model.

It cannot tell us what we ought to value.

That second question does not become less technical because it is philosophical.

PADLET • NO LOGIN • NO NAMES

Post one idea for each question.



Scan to open the Padlet.

- 1** What are your first-year goals? Academic or personal.
- 2** What are your life goals? Academic or personal.
- 3** What worries you about your first year at UBC?
- 4** How did you learn successfully in high school, and what might work at UBC?

TOMORROW

**We will begin with what you put
on the Padlet.**